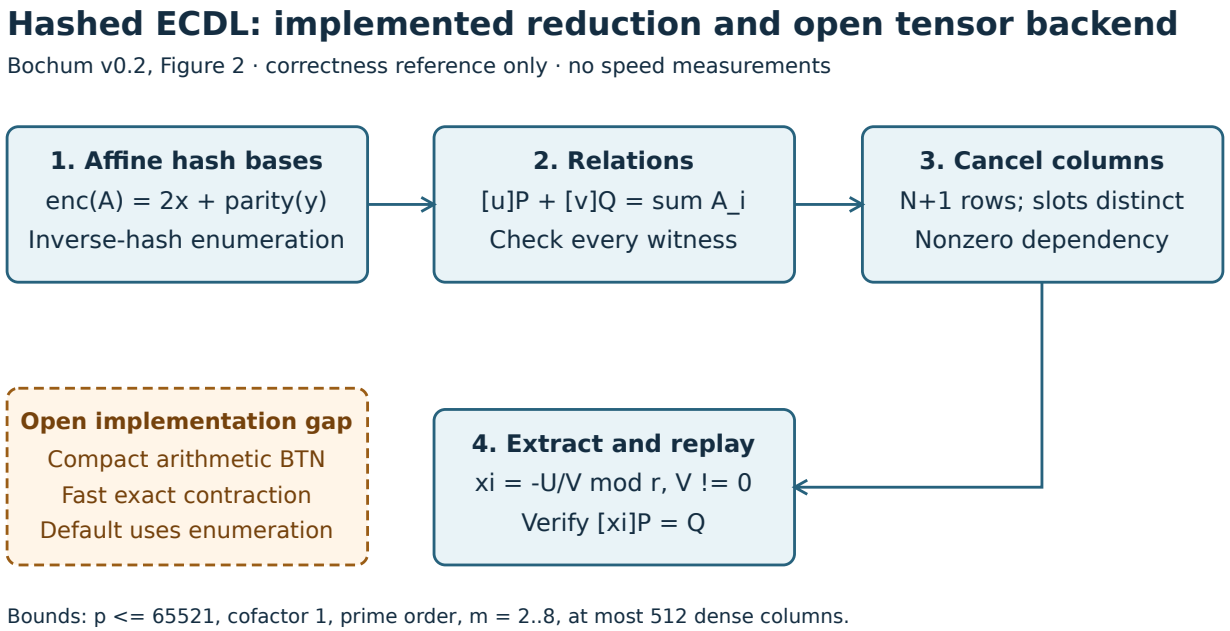
Hashed-factor-base ECDL reference

2026-10-29 · Implementation report · Partial replication

The requested source is Akshima et al., *A Subexponential Time Algorithm for Elliptic Curve Discrete Logarithm*, version 0.2, October 29, 2026. Upstream revision: `1ee44e562b1b9a50ea339f3b8f81112b5a24b96f`. Source repository.

Delivered code

The same native Rust algorithm is exposed as `crypto_lib::cryptanalysis::hashed_ecdl` and `cryptanalysis::hashed_ecdl`. It implements the affine-hashed bases of section 3.2, inverse-hash enumeration of section 3.5, and Figure 2′s relation/dependency reduction. Every decomposition is checked; a certificate retains the successful trial's hashes, relations and modular dependency. Replay checks every relation, cancellation, V != 0, and the final scalar multiplication.



Implementation boundary

The default oracle enumerates m-1 slots and checks the final group residual. It is generic decomposition, not tensor contraction. The compact O(mb)-edge, low-degree Boolean arithmetic/decomposition encoder and the assumed fast contraction algorithm are **not implemented**. The paper's Theorem 2 relies on external Theorem D.1 of [Ope26]; it was not independently verified here. The full conditional asymptotic replication remains unresolved.

The reference supports prime fields at most 65521, exact cofactor-one odd-prime cardinality, 2 through 8 slots and at most 512 dense columns. Public seeded SplitMix64 supplies reproducible fixtures, not a cryptographic or ideal uniform randomness guarantee. The small-field square-root table has O(p) setup and space. These bounds and choices do not implement the paper's full complexity.

Validation and remaining work

Nine shared unit tests cover independent point-scan factor-base checks, all-target oracle checks, every logarithm in a 19-point fixture group, seeded two/three-slot solves, corruption rejection, V=0 extraction and resource limits. The cryptanalysis API additionally cross-checks scalar arithmetic against its independent C implementation. Actual integration commands and outcomes are recorded in the PR; the Markdown companion gives reproducible commands.

No performance experiment, IC/rho comparison, deployed-curve run, new theorem, Lean proof or general runtime guarantee is claimed. Scoreboards, progress timelines, leaderboards and curve diagrams remain unchanged because no admitted measurement or new curve/route fact was produced. The open problem record `problems/hashed-ecdl-tensor-backend.md` in crypto names the missing encoder, backend, review and validation exit conditions. Passing tests or merging this code does not resolve that problem.

Sources checked

Upstream abstract, limitations, Theorems 2/4 and sections 3.2–3.5; crypto’s index-calculus boundary ledger, `semaev_decomp.rs` and `aps_single_relation.rs`; cryptanalysis’s existing multiplicative finite-field IC binding; and the sibling ECDLP atlas’s first-fall-degree assumption lead. No novelty or performance conclusion is inferred from that review.